

NATIONAL RESEARCH UNIVERSITY
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Faculty of Computer Science
Bachelor's Programme "Applied Mathematics and Informatics"

Research Project Report on the Topic:
Simulation Modeling of Market Shocks and their Consequences

Submitted by the Student:

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Contents

1	Annotation	3
1.1	На русском:	3
1.2	In English:	3
2	Keyword list	3
3	Introduction	4
4	Main chapters	5
4.1	Chapter 1: Problem Statement and Theoretical Analysis of Market Shocks	5
4.1.1	Purpose of the chapter:	5
4.1.2	Content:	5
4.1.3	Conclusion of the chapter:	6
4.2	Chapter 2: Modeling methodology and description of the simulator	6
4.2.1	Purpose of the chapter:	6
4.2.2	Content:	6
4.2.3	Conclusion of the chapter:	7
4.3	Chapter 3: Experimental study and analysis of results	7
4.3.1	Purpose of the chapter:	7
4.3.2	Content:	7
4.3.3	Conclusion of the chapter:	8
5	Conclusion	8
6	List of references	8

1 Annotation

1.1 На русском:

Данная работа посвящена исследованию моделирования рыночных шоков с использованием агентно-ориентированных моделей. В работе описывается постановка задачи – анализ поведения рыночной системы при возникновении внезапных изменений условий торговли. Основное внимание уделяется разработке экспериментов с использованием симулятора, реализованного в модуле AgentBasedModel, анализу влияния различных параметров (число трейдеров, рыночных мейкеров, интенсивность событий) на динамику системы и валидации результатов моделирования. Полученные результаты позволяют выявить закономерности формирования волатильности, изменения ликвидности и адаптивного поведения агентов в условиях рыночных потрясений. Работа имеет практическую значимость для разработки стратегий стабилизации финансовых рынков и оценки рисков в нестабильных экономических условиях

1.2 In English:

This work is dedicated to the study of market shock modeling using agent-based models. The paper outlines the research problem—analyzing the behavior of a market system under sudden trading condition changes. The primary focus is on developing experiments using a simulator implemented in the AgentBasedModel module, analyzing the impact of various parameters (such as the number of traders, market makers, and event intensity) on the system dynamics, and validating the simulation results. The obtained results reveal patterns in volatility formation, liquidity changes, and the adaptive behavior of agents under market stress. The work is of practical importance for developing strategies to stabilize financial markets and assess risks in unstable economic conditions

2 Keyword list

- 1 Market shocks
- 2 Agent-based modeling
- 3 Simulation
- 4 Financial markets
- 5 Liquidity

6 Volatility

7 Parametric analysis

8 Experimental study

3 Introduction

Financial asset markets are dynamic and complex systems where any sudden change in the terms of trade can lead to significant price fluctuations, reduced liquidity and even systemic crises. Such phenomena, referred to as market shocks, attract the attention of researchers and practitioners, as their consequences can have far-reaching effects on the economy as a whole. In the current conditions of globalization and digitalization of financial markets, the issue of modeling market shocks becomes particularly relevant.

The purpose of this study is to develop and apply agent-based simulation to analyze the behavior of the financial system when market shocks occur. The research uses a simulator implemented in the AgentBasedModel module to reproduce the dynamics of trading on exchanges involving traders, market makers and other agents. The objectives of the research include:

- 1 Studying the baseline behavior of the model without shocks to establish control dynamics.
- 2 Introducing different scenarios of market shocks (e.g., a sudden change in liquidity or a sharp decline in demand) and analyzing their impact on key system indicators.
- 3 Assessing the sensitivity of the model to changes in parameters such as the number of trading participants, intensity of shock events and the characteristics of market makers.
- 4 Comparison of the obtained results with theoretical expectations and empirical data presented in the literature.

The relevance of the study is determined by the growing need to develop effective forecasting and risk management tools in conditions of increased market volatility. The results of the work can be useful for the formation of strategies to stabilize financial markets and the development of early warning methods for systemic risks.

The structure of the paper is as follows: the section “Literature Review” includes the analysis of the main approaches and comparative analysis of existing methods of modeling market shocks. The main chapters contain a detailed description of the modeling methodology (Chapter 1), description of the experimental part and analysis of the obtained results (Chapter 2). The

conclusion summarizes the results of the study, discusses its achievements and prospects for further research.

4 Main chapters

4.1 Chapter 1: Problem Statement and Theoretical Analysis of Market Shocks

4.1.1 Purpose of the chapter:

To describe in detail the problem of modeling market shocks, justify the relevance of the study, formulate research hypotheses and determine the main parameters of the model.

4.1.2 Content:

1 Theoretical background of market shocks:

- 1.1. Definition of a market shock and its main characteristics (sudden change in price, liquidity, trading volume)
- 1.2. Review of the causes of shocks: external economic events, regulatory changes, technological failures
- 1.3. Analysis of the effects of shocks on financial markets based on empirical studies (see Danielsson, 2005; Brunnermeier, 2009)

2 The relevance of modeling market shocks:

- 2.1. The rationale for the need for modeling to predict and prevent systemic crises.
- 2.2. Advantages of the agent-based approach compared to traditional economic modeling techniques (Epstein Axtell, 1996; Tesfatsion, 2003).

3 Formulation of research objectives:

- 3.1. To investigate the dynamics of the model without shocks and establish baselines.
- 3.2. To introduce different scenarios of market shocks followed by comparative analysis.
- 3.3. Conduct parametric analysis of the influence of various factors (number of traders, intensity of events) on the behavior of the system.

4.1.3 Conclusion of the chapter:

Brief summary of the main points, formulated hypotheses and objectives of the research, preparation of ground for description of modeling methodology in the next chapter.

4.2 Chapter 2: Modeling methodology and description of the simulator

4.2.1 Purpose of the chapter:

To present the technical and methodological background of the study, to describe the architecture of the simulator used, and to justify the choice of parameters and experiment scenarios.

4.2.2 Content:

1 Description of the simulator:

- 1.1. AgentBasedModel/traders.py - all traders and their parameters
- 1.2. AgentBasedModel/simulator.py - Simulator object and its initialization
- 1.3. AgentBasedModel/exchange.py - Exchange objects, assets and their initialization
- 1.4. AgentBasedModel/extra/events.py - what events can be triggered on the stock exchange during trading
- 1.5. AgentBasedModel/visualization/ - built-in visualization tools

2 Customization of the experimental environment:

- 2.1. Description of the procedure for creating and setting up the environment (installing dependencies, setting up the Jupyter kernel).
- 2.2. Determination of initial values of model parameters and justification of their choice based on literature data (Lux Marchesi, 1999).

3 Experimental scenarios:

- 3.1. Formulation of the baseline scenario (model without exposure to shock events).
- 3.2. Design of experimental scenarios modeling market shocks (e.g., sudden decrease in liquidity, sudden change in demand).
- 3.3. Methodology of data collection and processing, and a description of the statistical techniques used (e.g. regression analysis, time series analysis).

4.2.3 Conclusion of the chapter:

Conclusions on the feasibility of the chosen modeling approach and methodology, preparation for the experimental study.

4.3 Chapter 3: Experimental study and analysis of results

4.3.1 Purpose of the chapter:

To conduct a series of experiments using the developed simulator, analyze the impact of market shocks on the system behavior and compare the results with theoretical expectations.

4.3.2 Content:

1 Description of the experiments:

- 1.1. Presentation of the basic simulation results in the absence of shock events.
- 1.2. Description of the experiments with the introduction of different shock scenarios:
 - 1.2.1. Sudden decrease in liquidity.
 - 1.2.2. A sudden increase in trading volume when the price falls.
- 1.3. Detailed description of changes in key parameters and methodology of experiments.

2 Data analysis:

- 2.1. Presentation of experimental results in the form of graphs, tables and statistical measures.
- 2.2. Comparison of the system behavior under the baseline scenario and scenarios with shocks.
- 2.3. Discussion of the identified patterns, comparing with the findings of previous studies (see Lux Marchesi, 1999; Samanidou et al., 2007).

3 Discussion of the results:

- 3.1. Interpretation of the results obtained in the context of the hypotheses posed.
- 3.2. Analysis of the limitations of the model and possible reasons for discrepancies with theoretical expectations.
- 3.3. Suggestions for further improvement of the model and research methodology.

4.3.3 Conclusion of the chapter:

Summary conclusions on the conducted experiments, evaluation of the effectiveness of agent-based modeling for the analysis of market shocks, formulation of recommendations for further research.

5 Conclusion

The conclusion summarizes the results of the research, evaluates the achieved results and formulates the prospects for further work. In the course of the study, the methodology of agent-based modeling was developed to analyze the behavior of the financial system under market shocks. The results obtained showed that:

- 1 Agent-based modeling effectively reproduces the dynamics of financial markets, demonstrating characteristic patterns under abrupt changes in conditions.
- 2 The introduction of shock events leads to significant changes in volatility and liquidity, confirming hypotheses derived from theoretical studies (Danielsson, 2005; Brunnermeier, 2009).
- 3 The parametric analysis allowed us to identify the key factors that influence the stability of the system, as well as to determine the scope of application of the model for the development of practical strategies for market stabilization.

The main limitations of the study are related to the simplified model of the market system, which requires further expansion of the simulator's functionality and deeper empirical substantiation. Prospects for further research include the integration of additional factors (e.g., the impact of regulatory measures, intermarket interactions) and the application of the model to the analysis of real market data.

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